Math 1200 Review Session 2



Daniel Salkinder

daniel.salkinder@yale.edu

Office hours: Fridays 4-5pm, WLH 015

(Mostly) Review

Sketch the function $f(x) = 2 - 2\cos(2x)$ from $x = 0$ to $x = 2\pi$.

What is the derivative?

Sketch the arc length (distance traveled along the curve) as a function of x.

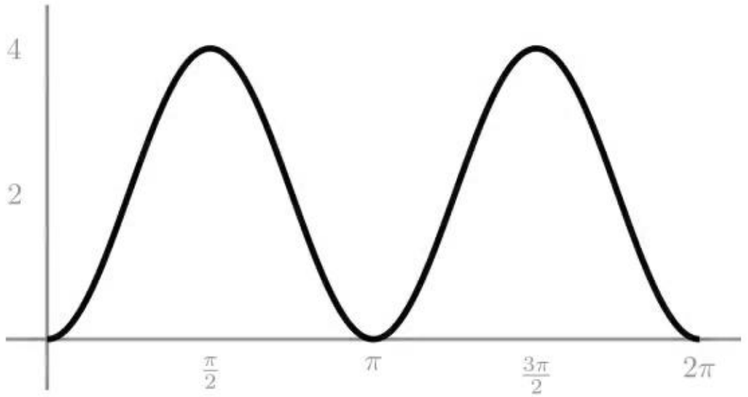
(Mostly) Review

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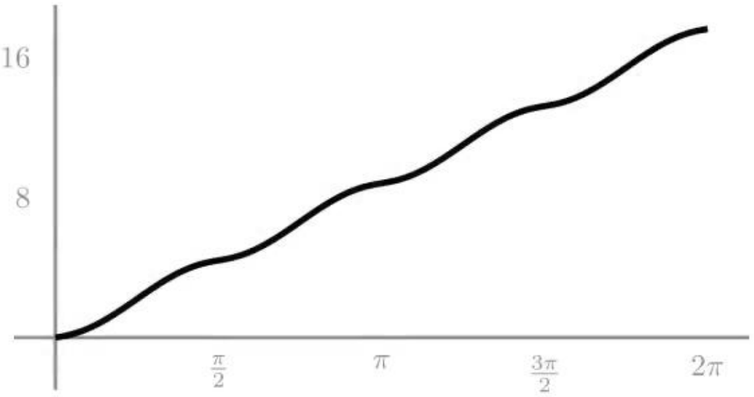
What is the derivative?

Sketch the arc length (distance traveled along the curve) as a function of x.

$y = f(x)$



arc length $s(x)$



$$f'(x) = 4\sin(2x)$$

2D Geometry

Let $f(x) = \sqrt{x}$.

Sketch the graph of $y = f(x)$.

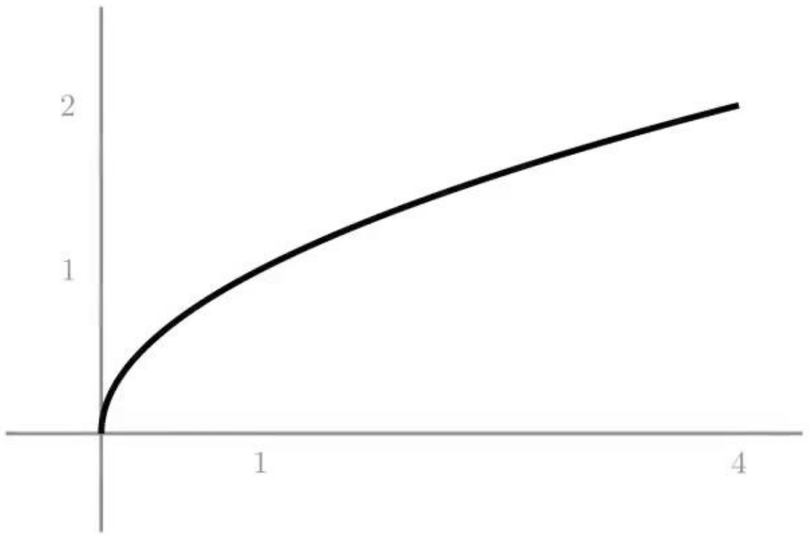
Sketch the graph of $x = f(y)$.

Let $f(x) = \sqrt{x}$.

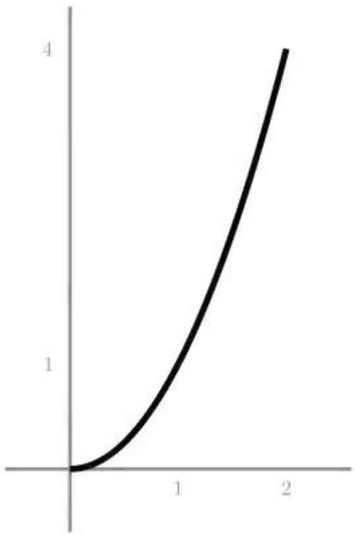
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$y = f(x)$



$x = f(y)$



The graph of $y = e^x$ is reflected across the line $y = x$. What do you get?

A) e^{-x}

B) $-e^x$

C) $\log(x)$

D) not a function

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The point $(1, 2)$ is rotated $90°$ clockwise about the origin. Where does it land?

A) $(2, 1)$

B) $(1, -2)$

C) $(-2, 1)$

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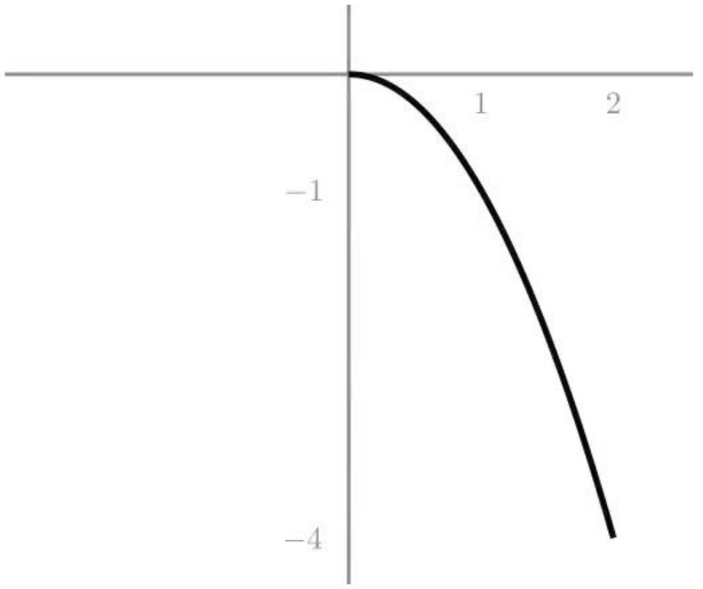
D) $(2, -1)$

Let $f(x) = \sqrt{x}$. If you rotate the graph of $f(x)$ $90°$ clockwise, what function does it trace out?

What are the domain and range of the function corresponding to the rotated graph?

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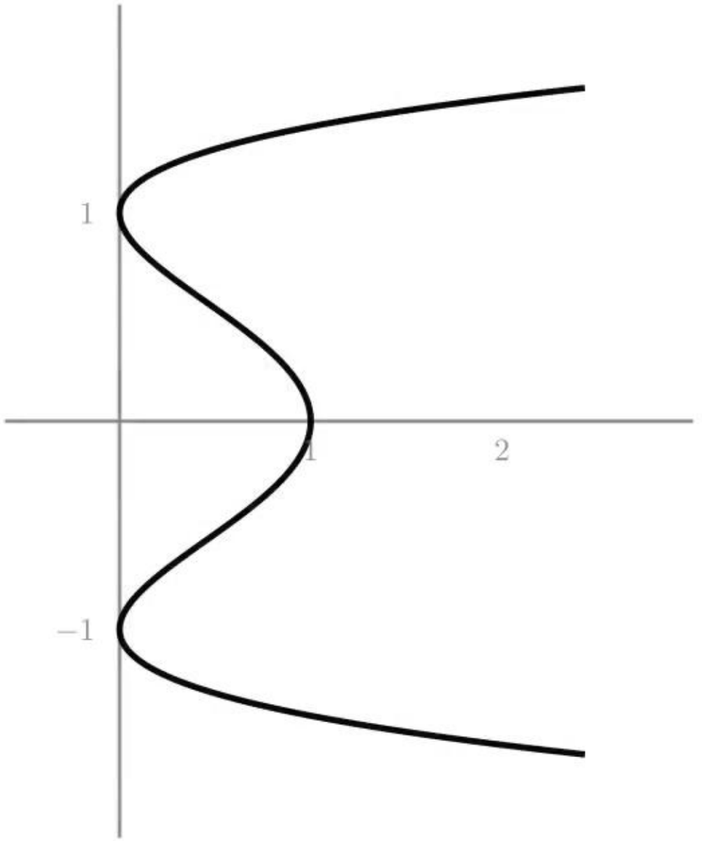
$y = -x^2$

domain: $x \geq 0$

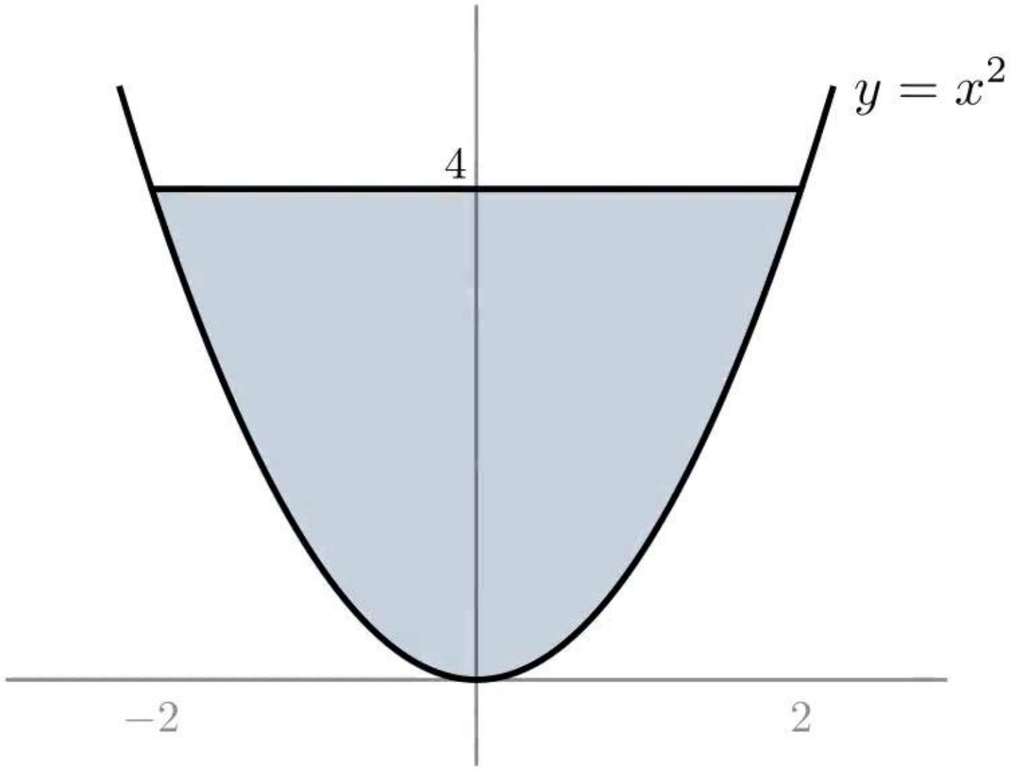
range: $y \leq 0$

Draw the graph of $x = (y+1)^2(y-1)^2$.

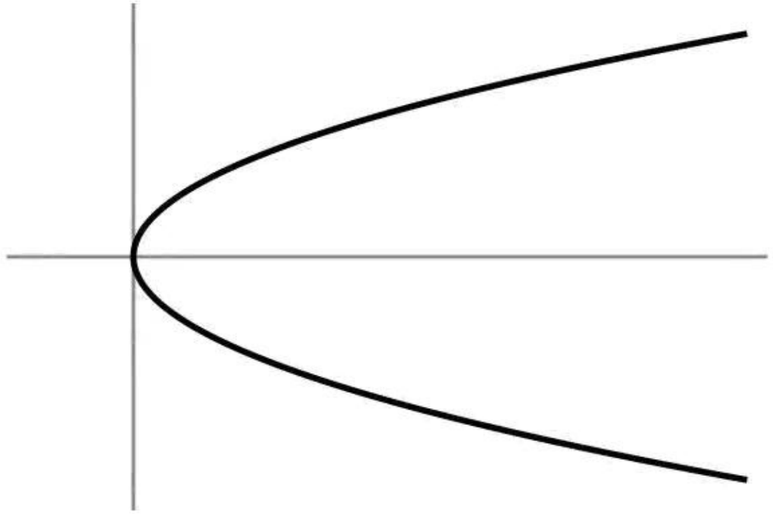
Draw the graph of $x = (y+1)^2(y-1)^2$.



Give two descriptions of the shaded region.



Consider the graph below ($x = y^2$). Which is true?



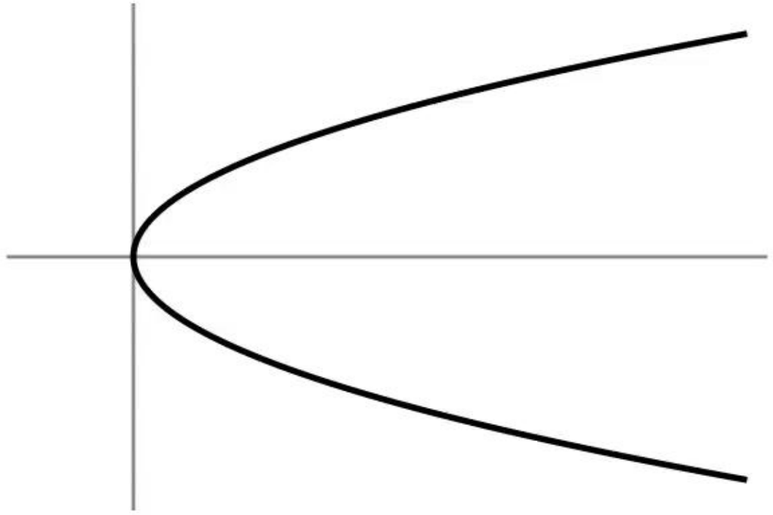
A) y is a function of x AND x is a function of y

B) y is a function of x AND x is NOT a function of y

C) y is NOT a function of x AND x is a function of y

D) y is NOT a function of x AND x is NOT a function of y

Consider the graph below ($x = y^2$). Which is true?



A) y is a function of x AND x is a function of y

B) y is a function of x AND x is NOT a function of y

C) y is NOT a function of x AND x is a function of y

D) y is NOT a function of x AND x is NOT a function of y

Same question for $x^2 + y^2 = 1$. Which is true?

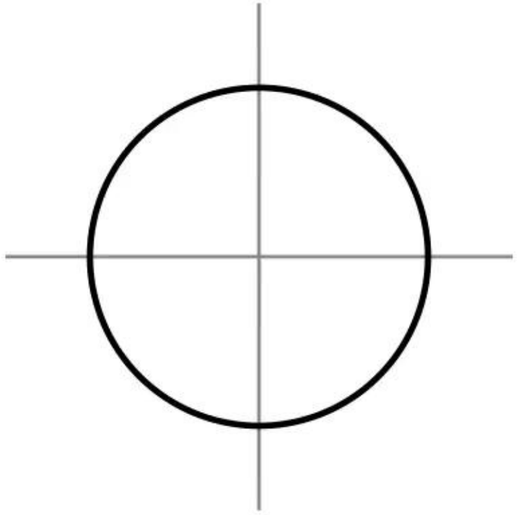
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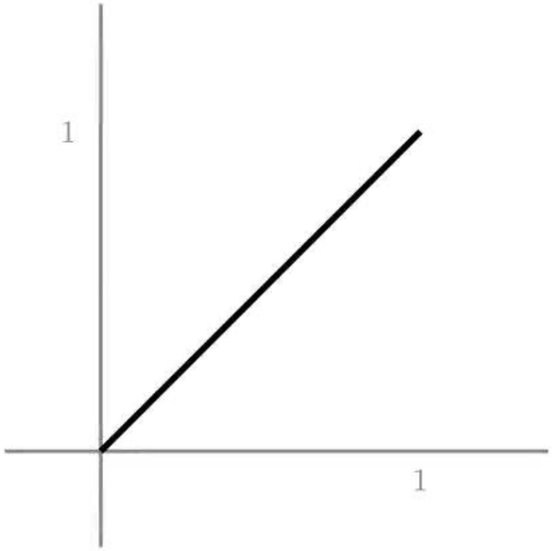
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Parametric Equations

Reparametrizations

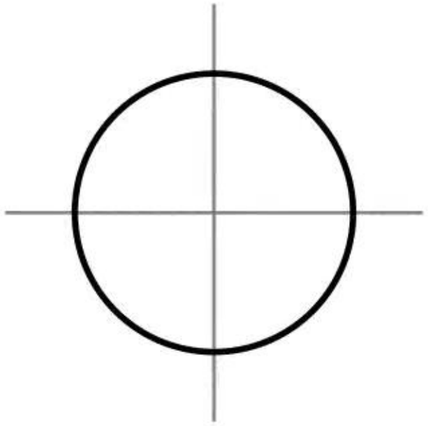


$$(x, y) = (u, u), \qquad 0 \leq u \leq 1$$

$$(x, y) = (e^{-t}, e^{-t}), \qquad 0 \leq t < \infty$$

Have we broken geometry?

Which of the following is NOT a parametrization of $x^2 + y^2 = 1$?



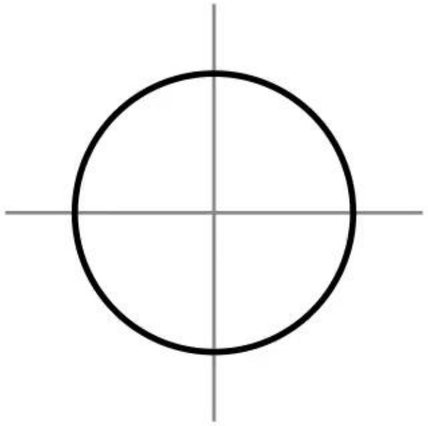
A) $(-\cos t,\ -\sin t), \quad 0 \le t \le 2\pi$

B) $(\sqrt{t},\ \sqrt{1-t}), \quad (-\sqrt{t},\ \sqrt{1-t}),$
$(-\sqrt{t},\ -\sqrt{1-t}), \quad (\sqrt{t},\ -\sqrt{1-t}), \qquad 0 \le t \le 1$

C) $\big(\cos(2\pi(1-t)),\ \sin(2\pi t)\big), \quad 0 \le t \le 1$

D) $\left(\cos\frac{t}{2},\ \sin\frac{t}{2}\right), \quad 0 \le t \le 2\pi$

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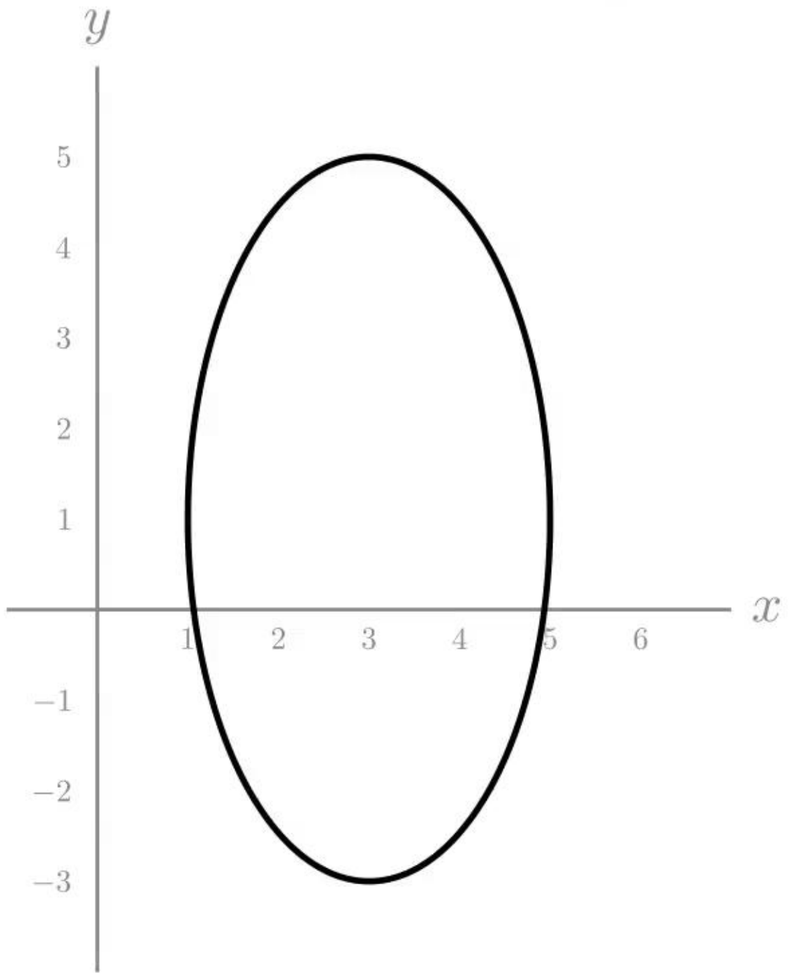
D) $\left(\cos\frac{t}{2},\ \sin\frac{t}{2}\right), \quad 0 \le t \le 2\pi$

Draw the curve parametrized by:

$$x = 3 - 2\cos(t) \qquad y = 1 + 4\sin(t), \qquad 0 \leq t \leq 2\pi$$

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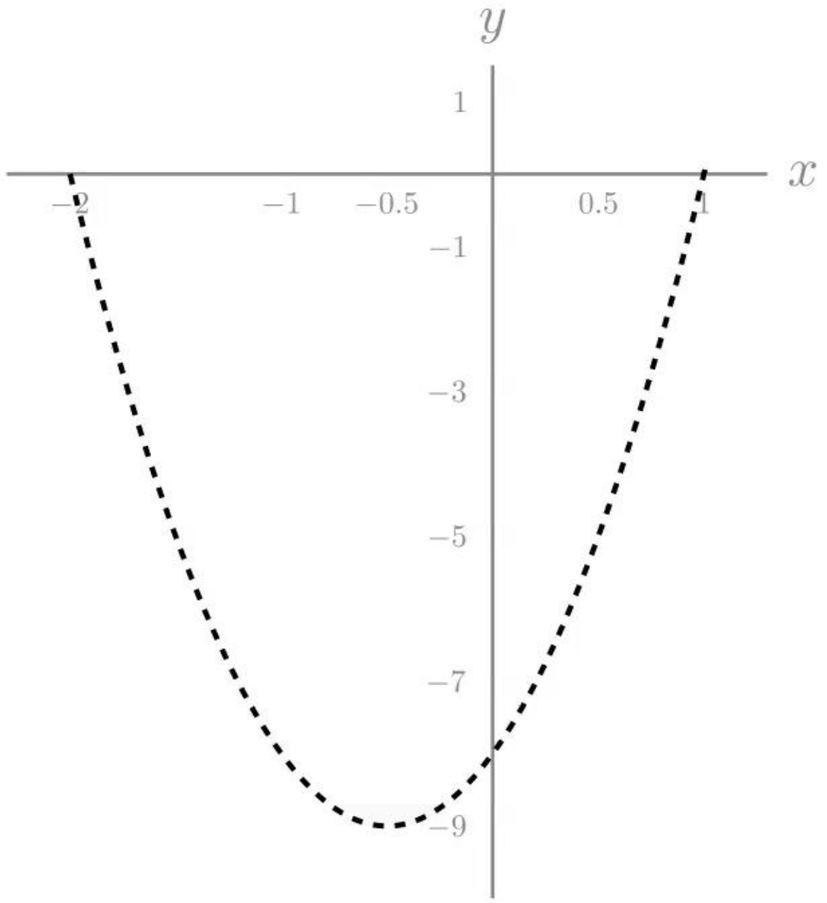


Draw the curve parametrized by:

$$x = \tfrac{1}{2}e^{-3t} \qquad y = e^{-6t} + 2e^{-3t} - 8$$

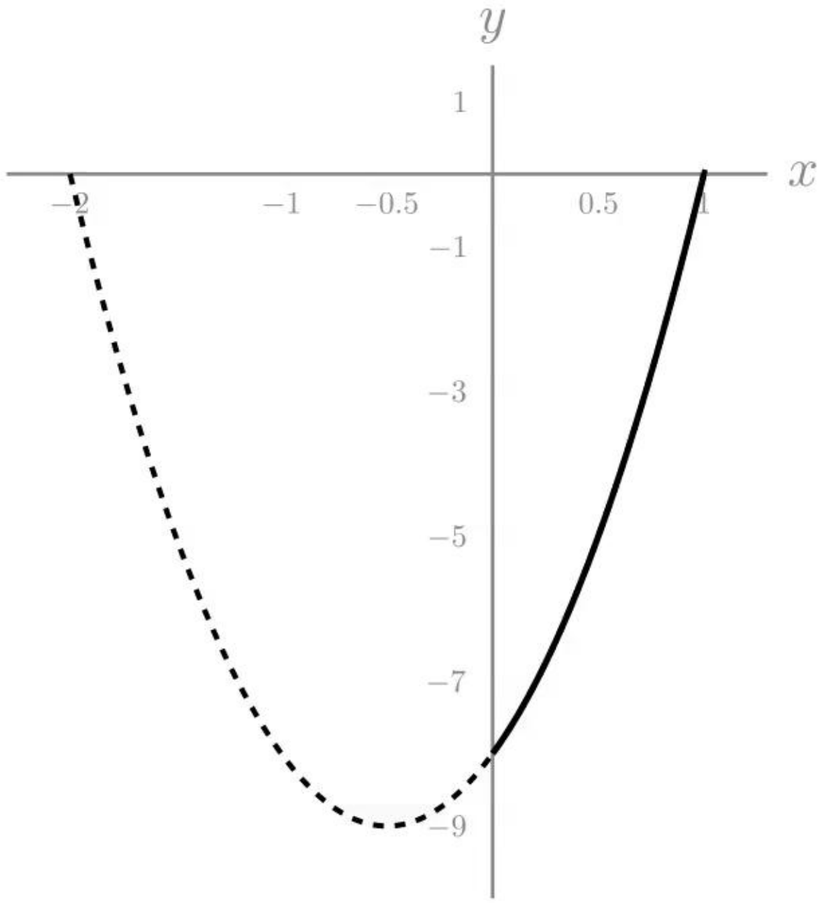
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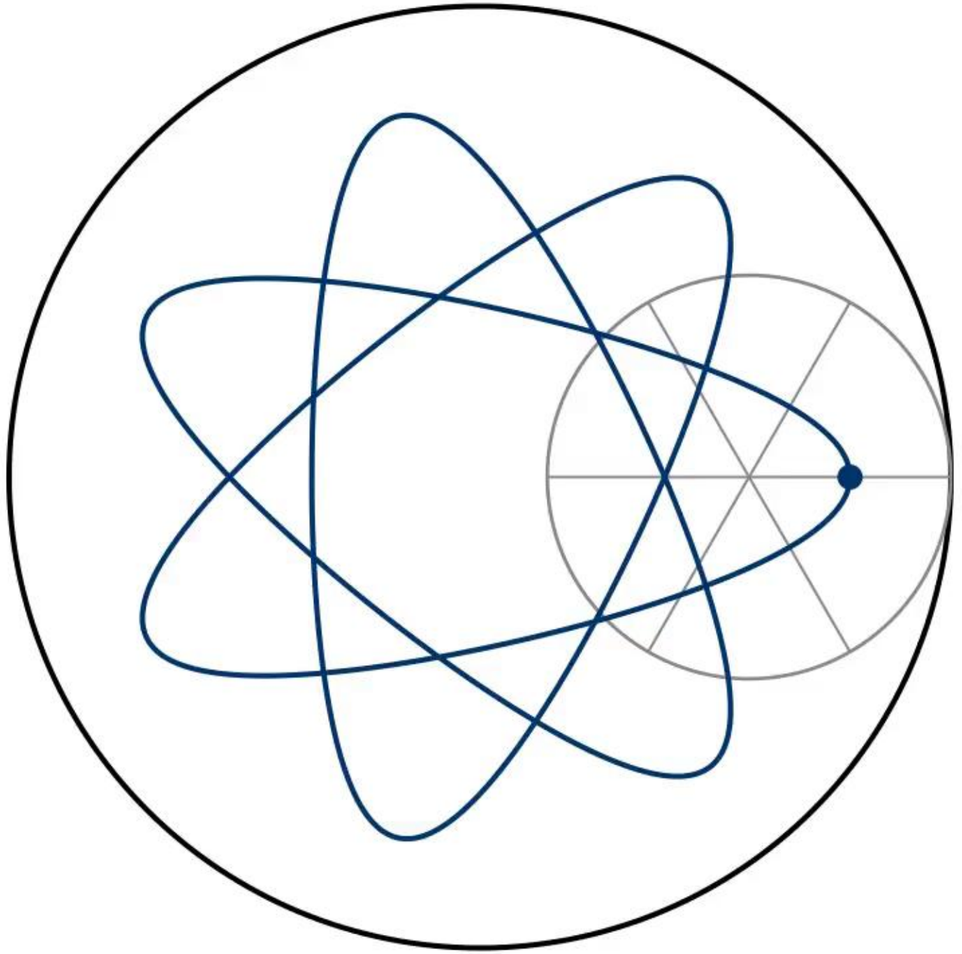
Spirograph



Spirograph


Iconic WHEELS AND RINGS
of the original spin art set,
updated for today!

Spirograph



Exit Poll

1) How do you feel about the pace of the course so far?

2) How do you plan on studying for your exam in 2 weeks?

3) How did you study for math tests before this class?

Spirograph in Desmos